

HisabDo AI Roadmap & Documentation

Department 3 – AI/ML Team

Project: HisabDo Website Enhancement

Deliverable: AI Roadmap & Documentation

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1. Executive Summary

The purpose of this AI Roadmap is to define a practical strategy for introducing Artificial Intelligence features into the HisabDo ecosystem.

The proposed AI strategy focuses on creating an intelligent and scalable **HisabDo AI Assistant** that can help users understand the platform, answer frequently asked questions, provide product guidance, improve user engagement, and support future AI-powered financial features.

The roadmap is designed around a phased approach.

The first phase focuses on building a controlled and reliable AI Assistant prototype.

Future phases can introduce more advanced capabilities such as:

- Personalized recommendations
- Financial insights
- Smart content suggestions
- Document processing
- OCR capabilities
- Retrieval-Augmented Generation (RAG)
- Intelligent analytics
- Multilingual support

The objective is not to add AI only as a marketing feature. The objective is to introduce AI capabilities that solve real user problems and provide measurable value to HisabDo users.

2. AI Vision for HisabDo

The long-term vision is to transform HisabDo from a traditional application into an intelligent financial assistance ecosystem.

The AI layer should help users:

- Understand HisabDo features
- Find relevant information quickly
- Receive product guidance
- Get answers to common questions
- Navigate the application easily
- Understand financial concepts
- Discover relevant features
- Receive personalized assistance in future versions

The AI system should be designed as a reusable service that can support:

Website



Web Application



Mobile Application



HisabDo AI Service

Instead of building separate AI systems for every platform, the recommended approach is to create one centralized AI service layer.

3. Finalized Chatbot / AI Assistant Use Case

Proposed Feature

HisabDo AI Assistant

The HisabDo AI Assistant will be an intelligent conversational support system designed to help users interact with the HisabDo ecosystem using natural language.

The first version should focus on practical and controlled capabilities.

Primary Purpose

The AI Assistant should help users with:

- Frequently asked questions

- Product information
 - Feature guidance
 - Navigation assistance
 - Basic financial education
 - Expense management guidance
 - General accounting-related guidance
-

4. Target Users

The AI Assistant can serve multiple types of users.

1. New Users

New users may not understand how HisabDo works.

Example:

How does HisabDo help me manage my business?

The AI Assistant can provide a clear explanation.

2. Existing Users

Existing users may need quick guidance while using the platform.

Example:

Where can I check my expenses?

The AI Assistant can guide the user toward the correct feature.

3. Small Business Users

Small business owners may need basic guidance about:

- Expenses
 - Transactions
 - Cash flow
 - Financial records
 - Business management
-

4. Website Visitors

Visitors may want to understand:

- What HisabDo is
 - Who it is for
 - What features are available
 - How the application works
-

5. Core AI Assistant Use Cases

5.1 FAQ Assistant

The AI Assistant can answer common questions.

Example:

User:

How does HisabDo work?

AI:

HisabDo helps users organize financial activities, manage expenses, track transactions, and access relevant financial information.

5.2 Product Guidance

The AI can explain product features.

Example:

User:

How can I manage my expenses?

AI:

You can manage expenses by recording transactions, categorizing spending, and reviewing financial activity through the available HisabDo features.

5.3 Navigation Assistance

The AI Assistant can help users locate relevant pages or features.

Example:

User:

Where can I view my reports?

AI:

You can access your financial reports from the
Dashboard or Reports section.

5.4 Basic Financial Education

The AI Assistant can provide simple educational information.

Example:

User:

What is cash flow?

AI:

Cash flow represents the movement of money into
and out of a business.

5.5 Related Question Suggestions

After answering a question, the system can recommend related questions.

Example:

Related Questions:

- How can I track expenses?
- What is monthly cash flow?
- How can I review financial reports?

This can improve user engagement.

6. AI Conversation Flow

The recommended AI conversation flow is:

User



User enters question



Input Validation



Question Classification



FAQ / Knowledge Base Search



AI Processing



Response Generation



Response Validation



Related Question Suggestions



User

7. AI Assistant Scope and Limitations

The first version of the AI Assistant should have a clearly defined scope.

Supported

- FAQ questions
- Product information

- Basic guidance
- Navigation support
- Accounting education
- General financial concepts

Not Supported

- Guaranteed financial predictions
- Legal advice
- Professional investment advice
- Unsupported sensitive decisions
- Fabricated information

The AI system should never invent information when it does not have sufficient knowledge.

Instead, it should respond with a safe fallback message.

Example:

I do not currently have enough verified information to answer this question accurately. Please check the relevant support section or contact the support team.

8. Current POC

The current Proof of Concept demonstrates that an AI/ML service can be structured as a practical backend service.

The current implementation approach includes:

- FastAPI-based backend service
- Structured API endpoints
- Input validation
- Response validation
- Error handling
- Health check endpoint
- Swagger API documentation
- Prototype UI

- API testing
- End-to-end workflow demonstration

The current AI POC provides the foundation for future AI Assistant development.

Current POC Architecture

Frontend / Prototype UI



FastAPI Service



Input Validation



AI Processing Logic



Response Generation



Response Validation



Frontend Display

Current API Structure

Recommended endpoint:

POST /api/v1/ai-assistant

Example request:

```
{  
  "message": "How can I manage my business expenses?"  
}
```

Example response:

```
{  
  "status": "success",
```



```
"answer": "You can manage expenses by recording and categorizing your transactions.",
"category": "expense_management",
"confidence": "high",
"related_questions": [
  "How do I track daily expenses?",
  "How can I review monthly spending?"
]
```

9. Improve AI POC

The next objective is to improve the existing AI Proof of Concept.

The POC should move from a basic demonstration toward a structured AI service.

Required Improvements

1. Structured API

The API should follow a versioned architecture.

Example:

/api/v1/ai-assistant

This makes future updates easier.

2. Input Validation

The system should validate:

- Empty messages
- Missing messages
- Invalid data types
- Extremely long messages
- Invalid request formats

Example:

Invalid Request



400 Bad Request



Clear Error Message

3. Response Validation

Every successful response should contain a consistent structure.

Example:

```
{  
  "status": "success",  
  "answer": "AI response",  
  "category": "FAQ",  
  "confidence": "high",  
  "related_questions": []  
}
```

4. Error Handling

The AI service should handle errors safely.

Possible errors include:

- Invalid input
- Missing data
- API failure
- AI processing failure
- Internal server errors

Example:

```
{  
  "status": "error",  
  "message": "AI service is temporarily unavailable."  
}
```

}

Internal technical details should not be exposed to users.

5. FAQ Knowledge Base

The AI Assistant should use a structured knowledge base.

Example categories:

FAQ Knowledge Base

|

|— General

|— Account

|— Dashboard

|— Expenses

|— Transactions

|— Reports

|— Payments

|— Support

□ 10. AI Testing

AI testing should validate both technical functionality and user experience.

10.1 Functional Testing

Test whether the AI performs expected actions.

Examples:

- FAQ question
- Product question
- Navigation question
- General accounting question

Expected result:

Relevant and understandable answer

10.2 Input Validation Testing

Test:

Empty Input

Whitespace Input

Invalid JSON

Missing Fields

Long Input

Invalid Data Types

The API should return controlled errors.

10.3 Response Validation Testing

Verify that every response contains required fields.

Example:

status

answer

category

confidence

related_questions

10.4 Unsupported Question Testing

Test questions outside the AI scope.

Example:

Can you predict the stock market tomorrow?

Expected response:

The AI should not fabricate an answer.

Instead, it should provide a safe fallback.

10.5 API Testing

API testing should include:

- Endpoint testing
- HTTP status codes
- Request validation
- Response structure
- Error responses
- Health checks

Tools:

Swagger UI

Postman

FastAPI TestClient

Pytest

10.6 User Experience Testing

Evaluate:

- Is the chatbot easy to find?
 - Is the input interface clear?
 - Are answers easy to understand?
 - Are responses fast enough?
 - Are suggested questions useful?
-

11. Required Integration

The AI system should eventually integrate with three major platforms.

Website Integration

The website can include an AI Assistant through:

- Floating chatbot button
- Dedicated AI support page

- FAQ Assistant
- Smart search
- Contextual help

Example:

Website Visitor



Ask AI



HisabDo AI API



Knowledge Base



Response

Web App Integration

The Web App can use AI for:

- Contextual support
- User guidance
- Feature explanation
- Workflow assistance
- Personalized help

The AI can understand the current application context.

Example:

Current Page:

Expense Dashboard

User:

How do I add an expense?

AI:

Provide context-specific instructions.

Mobile App Integration

The mobile application can integrate with the same centralized AI API.

Example:

Mobile App



HTTPS API Request



HisabDo AI Service



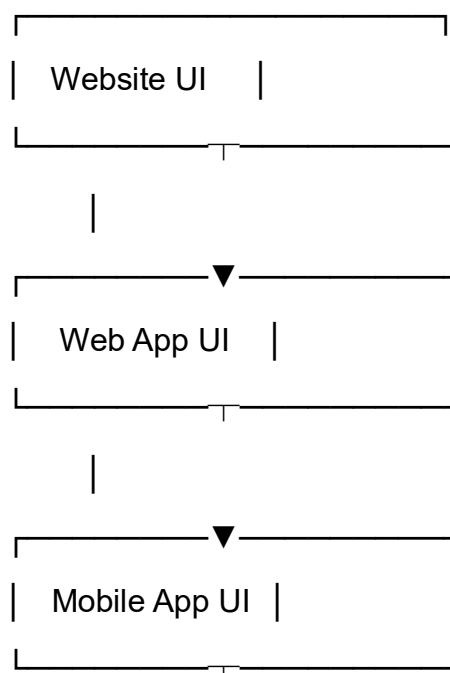
AI Response

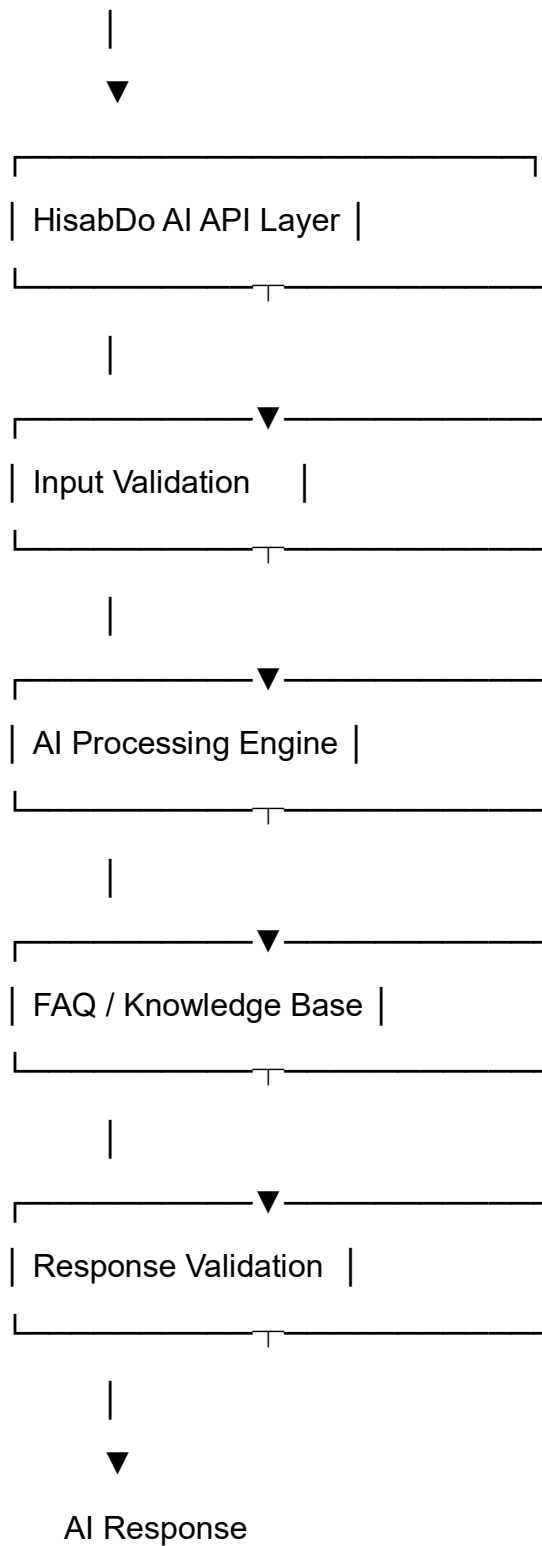


Mobile Chat Interface

The API should remain platform-independent.

12. Recommended Integration Architecture





13. Technical Requirements

The following technical requirements are recommended.

Backend

Python

FastAPI

Pydantic

Uvicorn

API Documentation

Swagger UI

OpenAPI

AI Processing

Future options:

LLM

RAG

Embeddings

Vector Database

Knowledge Base Retrieval

Database

Possible options:

PostgreSQL

MongoDB

MySQL

The selected database should depend on the existing HisabDo architecture.

Knowledge Base

The system should support:

- FAQs
- Product documentation
- Help articles

- Website content
 - User guides
-

Testing

Recommended tools:

Pytest

FastAPI TestClient

Postman

Swagger

14. Security Requirements

The AI system should implement:

- HTTPS
- API authentication
- Input validation
- Rate limiting
- Secure environment variables
- Error sanitization
- Logging
- Access control

Sensitive information should never be exposed directly through:

Frontend code

Error responses

Logs

Public APIs

15. Privacy Requirements

The AI system should follow a privacy-first approach.

Important principles:

- Collect only required data
 - Avoid unnecessary personal information
 - Avoid exposing financial information
 - Mask sensitive information in logs
 - Define data retention policies
 - Provide secure communication
-

16. Future Improvements

The AI Assistant can be improved in multiple phases.

Phase 1 — FAQ AI Assistant

Features:

- FAQ responses
 - Product guidance
 - Related questions
 - Knowledge-based answers
-

Phase 2 — Smart Recommendations

The system can recommend:

- Relevant content
- Helpful features
- Related guides
- Learning resources

Example:

Based on your activity,
you may find this guide useful.

Phase 3 — Personalized AI Assistant

Future AI can provide:

- User-specific guidance
 - Contextual recommendations
 - Personalized onboarding
 - Smart engagement
-

Phase 4 — RAG Architecture

The AI can use Retrieval-Augmented Generation.

Architecture:

User Question



Convert Question to Embedding



Search Vector Database



Retrieve Relevant Information



Send Context to AI Model



Generate Grounded Response

This can reduce hallucination and improve answer quality.

Phase 5 — OCR / Document Processing

Future AI can process:

- Receipts
- Invoices
- Financial documents

Example:

Upload Receipt



OCR Processing



Extract Text



Identify Amount



Identify Date



Identify Vendor



Create Expense Record

Phase 6 — AI Analytics & Insights

Future capabilities may include:

- Spending patterns
- Expense trends
- User behavior insights
- Anomaly detection
- Financial summaries

Example:

Your business expenses increased
by 15% compared to the previous month.

Phase 7 — Multilingual AI

Future support:

English

Urdu

Roman Urdu

Other Regional Languages

This can improve accessibility for Pakistani users.

17. AI Development Roadmap

Day 15 — AI Strategy

Tasks:

- Finalize AI Assistant use case
 - Define user problems
 - Define supported questions
 - Define scope and limitations
 - Prepare AI architecture
-

Day 16 — AI POC Improvement

Tasks:

- Improve AI prototype
 - Improve API structure
 - Add validation
 - Add error handling
 - Improve response format
 - Add knowledge-based responses
-

Day 17 — Testing and Integration

Tasks:

- Perform AI testing
- Test API
- Test invalid inputs
- Test responses
- Prepare integration architecture
- Document Website integration
- Document Web App integration

- Document Mobile App integration
-

18. AI Roadmap Summary

Phase 1

AI Assistant



Phase 2

FAQ + Knowledge Base



Phase 3

Recommendations



Phase 4

RAG Architecture



Phase 5

OCR / Documents



Phase 6

Analytics & Insights



Phase 7

Advanced Personalized AI

19. Final Deliverables

The AI/ML team should deliver:

AI_ROADMAP_AND_DOCUMENTATION.md

AI_FEATURE_RESEARCH.md

AI_CHATBOT_USE_CASE.md

AI_POC_DOCUMENTATION.md

AI_TESTING_REPORT.md

INTEGRATION_PLAN.md

API_DOCUMENTATION.md

Additional deliverables:

Architecture Diagram

AI Prototype

API Endpoints

Sample Inputs

Sample Outputs

Testing Results

Screenshots

GitHub Repository

Daily Progress Report

20. Conclusion

The HisabDo AI roadmap provides a structured path for introducing practical AI capabilities into the platform.

The first priority is a controlled and useful **HisabDo AI Assistant** that can provide:

- FAQ assistance
- Product guidance
- Navigation support
- Basic financial education

- Related question suggestions

The current POC demonstrates the technical foundation for AI integration through a structured API service.

The next steps focus on:

Improve AI POC



AI Testing



Integration Planning



Knowledge Base



Website Integration



Web App Integration



Mobile App Integration



Advanced AI Features

The long-term objective is to create an intelligent, scalable, secure, and user-focused AI ecosystem that can support the future growth of HisabDo.